

Types of Environment

1. Fully observable / Partially Observable
2. Deterministic / Stochastic
3. Episodic / Sequential
4. Single-agent / Multi-agent
5. Static / Dynamic
6. Discrete / Continuous

Fully observable / Partially Observable

Fully observable

- An agent can always see the entire state of an environment.
- Example: Chess



Partially Observable

- An agent can never see the entire state of an environment.
- Example: Card game



Deterministic / Stochastic

Deterministic

- An agent's current state and selected action can completely determine the next state of the environment.
- Example: Tic tac toe



Stochastic

- A stochastic environment is random in nature and cannot be determined completely by an agent.
- Example: Ludo (Any games that involve dice)



Episodic / Sequential

Episodic

- Only the current percept is required for the action
- Every episode is independent of each other
- Example: part picking robot



Sequential

- An agent requires memory of past actions to determine the next best actions.
- The current decision could affect all future decisions.
- Example: Chess



Single-agent / Multi-agent

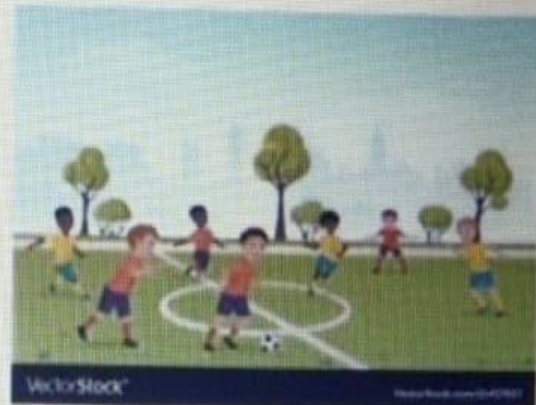
Single-agent

- If only one agent is involved in an environment, and operating by itself then such an environment is called single agent environment.
- Example: maze



Multi-agent

- if multiple agents are operating in an environment, then such an environment is called a multi-agent environment.
- Example: football



Static / Dynamic

Static

- The environment does not change while an agent is acting
- Example: Crossword puzzles



Dynamic

- The environment may change over time
- Example: roller coaster ride

Taxi driving



Discrete / Continuous

Discrete

- The environment consists of a finite number of actions that can be deliberated in the environment to obtain the output.
- Example: chess



Continuous

- The environment in which the actions performed cannot be numbered ie., is not discrete, is said to be continuous.
- Example: Self-driving cars



Known / Unknown

Known

- In a known environment, the results for all actions are known to the agent.
- Example: Card game



Unknown

- In unknown environment, agent needs to learn how it works in order to perform an action.
- Example: A new video game

